Structure & Syllabus for Semester-V

Integrated M.Sc. (Computer Science) Programme offered in

Department of Computer Science

Gujarat University

2025 – 2026

As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR UNDERGRADUATE PROGRAMMES, UGC

&

Resolution No. KCG/admin/2023-24/0607/kh.1

of

Education Department, Govt. of Gujarat

STRUCTURE FOR SEMESTER - 5 COURSES IN GUJARAT UNIVERSITY FOR INTEGRATED M.Sc. (COMPUTER SCIENCE) AS PER NEP 2025 - 2026

COURSE:	**INTEGRATED MASTER OF SCIENCE (COMPUTER SCIENCE)**
MAJOR:	**COMPUTER SCIENCE**
MINOR: (Any One Track)	
	1. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (AIML) **2. INFORMATION SECURITY (IS)** **3. WEB TECHNOLOGIES (WT)**

SEMESTER - 5

	MAJOR	
CODE	**COURSE**	**CREDITS**
DSC-C-CS-351T	MACHINE LEARNING	4
DSC-C-CS-352T	JAVA PROGRAMMING	4
DSC-C-CS-353P	JAVA PROGRAMMING – PRACTICALS	4
	MINOR	
CODE	**COURSE**	**CREDITS**
DSC-M-CS-354T	DATA ANALYTICS	2
DSC-M-CS-354P	DATA ANALYTICS – PRACTICALS	2
DSC-M-CS-355T	ARTIFICIAL INTELLIGENCE	2
DSC-M-CS-355P	ARTIFICIAL INTELLIGENCE – PRACTICALS	2
	OR	
DSC-M-CS-354T	CRYPTOGRAPHY	2
DSC-M-CS-354P	CRYPTOGRAPHY – PRACTICALS	2
DSC-M-CS-355T	BLOCKCHAIN TECHNOLOGY	2
DSC-M-CS-355P	BLOCKCHAIN TECHNOLOGY – PRACTICALS	2
	OR	
DSC-M-CS-354T	WEB SECURITY	2
DSC-M-CS-354P	WEB SECURITY - PRACTICALS	2
DSC-M-CS-355T	ADVANCED WEB SCRIPTING	2
DSC-M-CS-355P	ADVANCED WEB SCRIPTING - PRACTICALS	2
	INTER DISCIPLINARY / MULTI DISCIPLINARY	
CODE	**COURSE**	**CREDITS**
	NONE	
	ABILITY ENHANCEMENT COURSE	
CODE	**COURSE**	**CREDITS**
	NONE	
	SKILL ENHANCEMENT COURSE (ANY ONE COURSE)	
CODE	**COURSE**	**CREDITS**
SEC – 356	ADVANCED PYTHON PROGRAMMING – PRACTICALS	2
	COMMON VALUE-ADDED COURSES	
CODE	**COURSE**	**CREDITS**
	NONE	
	TOTAL CREDITS	**22**

5 Year Integrated MSc (Computer Science)

Exam Scheme for Semester – 5 (As per NEP)

Code	Course	Credits	Theory (Internal)	Theory (External)	Practical (Internal)	Practical (External)	Total
DSC-C-CS-351T	MACHINE LEARNING	4	50	50	***	***	100
DSC-C-CS-352T	JAVA PROGRAMMING	4	50	50	***	***	100
DSC-C-CS-353P	JAVA PROGRAMMING – PRACTICALS	4	***	***	50	50	100
DSC-M-CS-354T	DATA ANALYTICS	2	25	25	***	***	50
DSC-M-CS-354P	DATA ANALYTICS – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	ARTIFICIAL INTELLIGENCE	2	25	25	***	***	50
DSC-M-CS-355P	ARTIFICIAL INTELLIGENCE – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-354T	CRYPTOGRAPHY	2	25	25	***	***	50
DSC-M-CS-354P	CRYPTOGRAPHY – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	BLOCKCHAIN TECHNOLOGY	2	25	25	***	***	50
DSC-M-CS-355P	BLOCKCHAIN TECHNOLOGY – PRACTICALS	2	***	***	25	25	50
DSC-M-CS-354T	WEB SECURITY	2	25	25	***	***	50
DSC-M-CS-354P	WEB SECURITY - PRACTICALS	2	***	***	25	25	50
DSC-M-CS-355T	ADVANCED WEB SCRIPTING	2	25	25	***	***	50
DSC-M-CS-355P	ADVANCED WEB SCRIPTING – PRACTICALS	2	***	***	25	25	50
SEC – 356	ADVANCED PYTHON – PRACTICALS	2	***	***	25	25	50
TOTAL		22	150	150	125	125	550

Semester - V

(Integrated M.Sc. (Computer Science) Programme)

MAJOR: COMPUTER SCIENCE

GUJARAT UNIVERSITY

INTEGRATED M.Sc. (COMPUTER SCIENCE)

PROGRAMME OUTCOMES

- **PO1:** To inculcate self-learning and life-long learning capabilities
- **PO2:** To become an employable and successful entrepreneur
- **PO3:** To create skilled human resources in the area of computer science that caters the need of software development and system analysts
- **PO4:** To create skilled human resources that can work as software developers
- **PO5:** To empower the students with efficient communication skills for project management
- **PO6:** To inculcate capabilities like modelling business problems and business logic into software systems
- **PO7:** To inculcate leadership and teamwork qualities
- **PO8:** To inculcate qualities that helps to develop sustainable solutions to interdisciplinary global problems through research and innovation capabilities
- **PO9:** To promote independent and collaborative work in the area of computer applications
- **PO10:** To provide fundamental knowledge of computing

PROGRAMME SPECIFIC OUTCOMES

- **PSO1:** To inculcate and apply latest technologies to solve business problems
- **PSO2:** To apply algorithmic approach and computer science techniques in designing computer-based solutions and applications
- **PSO3:** To apply technical knowledge for developing secure, smart and sustainable solutions
- **PSO4:** To apply the knowledge of software engineering in designing effective business solutions
- **PSO5:** To be able to create, maintain and troubleshoot the business solutions
- **PSO6:** To be able to enhance their skills through life-long learning through professional activities
- **PSO7:** To develop ability for effective communication with the team members and management
- **PSO8:** To develop ability to work in multidisciplinary area as a team member and to develop leadership qualities and managerial skills
- **PSO9:** To develop effective solutions using technologies that can scale and provide high performance solutions
- **PSO10:** To exhibit professional and personal ethics in the industry

Course Name: Machine Learning

Course Code: DSC-C-CS-351T

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Develop a strong theoretical foundation for understanding of the state of art Machine Learning Algorithms
- **CO2:** Identify, formulate and solve machine learning problems that arise in practical applications
- **CO3:** Explore various tools and technologies that enable Machine Learning

Prerequisites:

Mathematical Foundations, Programming in C++/Python

Contents:

Unit No.	Particulars	Credits	Time
1.	**Unit Title: Overview of Machine Learning** Introduction to Machine learning from data, *Types of Machine Learning:* Supervised, Unsupervised, Reinforcement Learning, Concepts of Regression, Classification, Clustering	**01**	**15 Hrs.**
2.	**Unit Title: Linear Regression & Logistic Regression** Scatter diagram, Model representation for single variable, Single variable Cost Function, Least Square line fit, Normal Equations, Gradient Descent method for Linear Regression, Assumptions in Linear Regression, Properties of Regression line, Model Performance through R2, Multivariable model representation, Multivariable cost function, Multiple Linear Regression, Overfitting, Underfitting, Bias and variance, Regularization, Issues of using Linear Regression in Classification, Sigmoid function, Odds of an event, Logit function, Decision Boundary, Maximum likelihood function, Linear Regression verses Logistic Regression, Pros and Cons of Logistic Regression, Cost function, Multi-classification, Confusion Matrix, *Statistical measures to measure Classification*: Recall, Precision, Accuracy	**01**	**15 Hrs.**
3.	**Unit Title: Supervised Learning** Classification problems, Decision Boundaries, K-nearest Neighbour methods, Linear Classifiers, Bayes' Rule and Naive Bayes Model, SVM - Introduction, Support Vectors & Margin, Optimization Objective, Dimensionality Reduction, PCA, Linear & Non-Linear SVM, Hard Margin & Soft Margin, Large Margin Classifiers, Kernels, Decision Tree, *Ensemble methods for Classification and Regression:* Bagging, *Boosting:* AdaBoost, Random Forest	**01**	**15 Hrs.**

Unit No.	Particulars	Credits	Time
4.	**Unit: Unsupervised learning** Cluster Analysis, Classification and Clustering, Definition of Clusters, Clustering Applications, Distance Measures, Proximity Measures for Discrete Variables, Proximity Measures for Mixed Variables, Partitional Clustering, Clustering Criteria, K-Means Algorithm, Fuzzy Clustering, Hierarchical Clustering, Agglomerative Hierarchical Clustering, Divisive Hierarchical Clustering, Cluster validity, External criteria, Internal criteria	01	15 Hrs.

References:

1. Mitchell T., "Machine Learning", McGraw Hill Publications
2. Grus J., "Data Science from Scratch", O'Reilly Publications
3. Alpaydin E., "Introduction to Machine Learning", MIT Press
4. Coelho R., "Building Machine Learning Systems with Python", Packt Publishing Ltd.
5. Marsland S., "MACHINE LEARNING: An Algorithmic Perspective", CRC Press
6. Wunsch D., Xu R., "Clustering", IEEE Press
7. Ethem Alpaydın, "Introduction to Machine Learning", The MIT Press

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Develop an appreciation for what is involved in learning models from data
- Understand a wide variety of learning algorithms
  Understand how to evaluate models generated from data
- Understand and develop application involving computer vision and Natural Language Processing

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Course Name: Java Programming

Course Code: DSC-C-CS-352T

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Understand the concepts of Object-Oriented Programming Language and using Java
- **CO2:** Get good understanding of developing multi-user applications using Java Programming Language
- **CO3:** Get good understanding of developing multi-threaded applications using Java Programming Language
- **CO4:** Harness the features of Java using APIs of Collection Framework, Lambda expressions and streams for effective programming

Prerequisites:

Basic knowledge of C/C++ programming language

Contents:

Unit No.	Particulars	Credits	Time
1.	**Unit Title: Introduction, Data Type and Control Structures** *Introduction:* Recap of Object Oriented Programming Features and Relating to Java, Features of the Java Language, Java Environment, Object Oriented Programming in Java, Java Program Structure, Java and Unicode, Writing First Java Application, Command-Line Arguments, Use of System.out, System.console().printf and System.console().readline() Methods *Data Type, Variables and Constants, Loops and Logic & Exceptions:* Primitive Types, Reference Types, Difference from C++ in Usage of Reference Types, Arithmetic Calculations, Mixed Arithmetic Expressions, Mathematical Functions and Constants, Bitwise Operators, Enumerated Data Type, Program Comments, Loops, Logic and Scope, Making Decisions, Logical Operators, The Conditional Operator, The Switch Statement *Exceptions:* Understanding Exceptions, Types of Exceptions, Throw and Throws Keywords, Checked and Unchecked Exceptions, Handling Exceptions using Try-Catch and Finally Blocks, Multi-Catch Blocks, Creating Custom Exceptions	**01**	**15 Hrs.**
2.	**Unit Title: Defining Classes, Extending classes and Inheritance, Interface** Significance of Class, Defining Classes, Various Members for (which go in a) Class Definition – Instance Variable, Class Variable, Methods, Static Methods, Constructors, Initializer Block and Class Initializer Block, Method and Constructor Overloading, Using Objects, Use of *this* Keyword, Nested Classes, The Finalize() Method, Recap of Inheritance, Use of Keywords Extends and Super, "Is-A" Relationship for Inheritance, Packages and Access Specifiers, Package Keyword, Import and Import Static Keywords, Private, Public And Protected Keywords, Abstract Classes, Abstract Methods, Multiple Inheritance using Interfaces, Static and Default Methods in	**01**	**15 Hrs.**

Unit No.	Particulars	Credits	Time
	Interfaces, Functional Interfaces and Syntax of Lambda Expression		
3.	**Unit Title: Commonly used Classes, Packages: Custom Packages, In-built Packages** Commonly used Classes like String, StringBuilder, Math and Wrapper Classes, Creating and Importing Custom Packages, The java.io Package, File Class, IO Stream Classes for Reading and Writing to Files, RandomAccessFile Class, Stream API from java.util.stream Package, Stream, IntStream, LongStream, DoubleStream, Understanding the Intermediate and Terminal Operations on Streams, *reduce* and *collect* methods, Understanding the Collector, Creating useful Collector using methods from the Collectors Class LocalDate, LocalTime, LocalDateTime, OffsetDate, ZonedDateTime, LongSummaryStatistics, from java.util Package, Understanding the Optional Class methods for *classification and regression:* Bagging, Boosting, Random Forest	01	15 Hrs.
4.	**Unit: Threads And Collection Framework** Understanding Threads, Thread Class, Creating Thread of Executing using Runnable and using Sub-Class of Thread, Various Properties of Thread, using Synchronized Keyword, Using *wait* and *notify*, Collection Framework, Recap of Data Structures, The Collection Framework from java.util Package, Collection, Iterable, List, Set, Map, Queue, Dequeue, Utility Methods from Collections Class, Introduction to IDE	01	15 Hrs.

References:

1. Herbert Schildt, "Java, A beginners' guide", Oracle Press
2. Jain Pravin, "The class of Java", Pearson India
3. Horton I., "Beginning Java", 7th edition, Wrox India
4. Cay S Horstmann, Gary Cornell, "Core Java, Volume 1 – Fundamentals", Pearson Education
5. Ken Arnold, James Gosling, David Holmes, "The Java Programming Language", Addison-Wesley Pearson Education
6. Subramian, "Functional Programming in Java", VPragmatic Bookshelf
7. Urma R., Fusco M., Mycroft A., "Java 8 in action", Manning Publications

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Create appropriate classes using the Java Programming Language to solve a problem using Object Oriented Approach and use concepts of Functional programming
- Develop multi-user applications using Java Programming Language
- Create appropriate classes using Java Programming Language to solve a problem using Object Oriented approach and use concepts of functional programming
- Develop multi-threaded applications using the Java Programming Language
- Use data structures effectively to design applications in Java Programming Language

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Course Name: Java Programming – Practicals

Course Code: DSC-C-CS-353P

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Understand the concepts of Object-Oriented Programming Language and using Java
- **CO2:** Get good understanding of developing understanding of developing platform independent applications using java
- **CO3:** Get good understanding of multi-threaded applications using Java Programming language

Prerequisites:

Basic knowledge of C/C++ programming language

Contents:

Unit No.	Particulars	Credits	Time
1.	**Unit Title: Basics of Java Programming** Using Integrated Development Environment for Java Programming, Compiling & Executing Java Programs, Using Different Data Types in Java, Using various Operators & Performing Arithmetic Calculations, Decision Making with Java Programming, Looping, Implementation of Assertions, Use of Classes in Java, Constructors in Java, Exception Handling	01	30 Hrs.
2.	**Unit Title: Inheritance, Interface and Commonly used Classes** Implementation of Inheritance using Classes & Interfaces, Implementing Lambda, Implementation & Use of Packages in Java, Commonly used Classes & Methods	01	30 Hrs.
3.	**Unit Title: File Handling, Collection Framework** File Handling using Different Techniques like Input Output Streams: File.io, File.nio, Byte Streams and Buffered Input/Output, Reading, Writing and Creating Files; *Collection Framework: Interfaces for Collection:* Collection (Set, List, Queue, Deque) Map Interface	01	30 Hrs.
4.	**Unit: Threads** Processes and Threads, Implementing/Extending Thread Class & Runnable Interface in Java, Thread Objects, Pausing Execution, Join, Synchronization	01	30 Hrs.

References:

1. Herbert Schildt, "Java, A beginners' guide", Oracle Press
2. Jain Pravin, "The class of Java", Pearson India
3. Horton I., "Beginning Java", 7^{th} edition, Wrox India
4. Cay S Horstmann, Gary Cornell, "Core Java, Volume 1 – Fundamentals", Pearson Education

5. Ken Arnold, James Gosling, David Holmes, “The Java Programming Language”, Addison-Wesley Pearson Education
6. Subramian, “Functional Programming in Java”, VPragmatic Bookshelf
7. Urma R., Fusco M., Mycroft A., “Java 8 in action”, Manning Publications

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Create appropriate Classes using Java Programming Language to solve a problem using Object Oriented approach and use concepts of functional programming
- Develop multi-threaded applications using Java Programming Language
- Use data structures effectively to design applications in Java Programming Language

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